Decoder-Free Optical Learning for Axial Structure Recovery in Fixed-Focus Fluorescence Microscopy

Anonymous CVPR submission

Abstract

Recovering fluorescent structures that overlap in projection requires measurements that distinguish their axial arrangements. We present decoder-free optical learning for fixed-focus fluorescence microscopy. An angle-scanned plane wave illuminates a fixed, defocused phase plate, producing depth-dependent excitation while the specimen and detection focus remain stationary. We learn the plate by minimizing weighted squared correlations between recorded voxel responses at different depths and the same lateral position. Training uses the optical forward model, without specimen examples or a reconstruction decoder. We assess axial structure through intensity-profile recovery and controlled filament separation. On simulated mouse pyramidal neuron measurements, the learned plate reduces axial-profile Wasserstein error by 28–50 % and centroid error by 30–54 % relative to its random initialization. These gains hold across total variation and two plug-and-play methods at eightfold measurement compression and matched nominal excitation dose. The measurements also better distinguish neuronal references from merged-depth alternatives with the same lateral projection and column centroids, and crossing separation improves under two of the three priors. Under matched training conditions, the proposed criterion yields 13–20 % lower profile error than the tested coherence and depth-Fisher alternatives. The results demonstrate improved axial recovery across the three tested priors without coupling optical learning to a decoder.

1. Introduction

Fluorescent branches that overlap in a camera image may lie at different depths. Recovering their axial arrangement requires measurements that distinguish those depths, particularly when detection has a broad depth of field. If different arrangements produce similar observations, reconstruction relies more heavily on prior assumptions [2, 7]. Even an accurate mean depth can conceal two layers reconstructed as one. We ask whether learning the excitation illumination can improve the recovery of axial structure without choosing a reconstruction method during optical training.

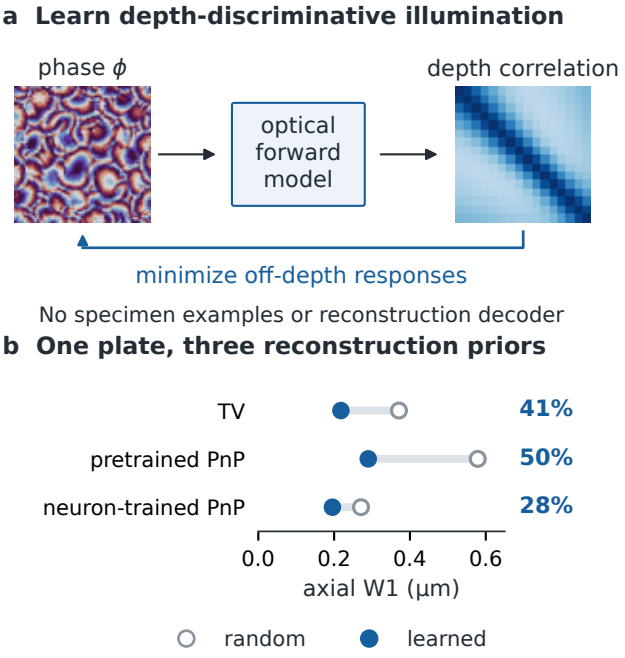


Figure 1. **Learn the illumination, then fix the plate for reconstruction.** (a) Optical training reduces correlations between depth-dependent measurement signatures. (b) The same learned plate improves axial-profile recovery with three reconstruction priors at eightfold measurement compression and matched nominal excitation dose. Means are over 60 neuronal crops; intervals are reported in Tab. 1.

Our system scans the incident angle of a plane wave through a fixed, defocused phase plate (Fig. 2). Propagation produces structured excitation whose lateral displacement under the scan varies with depth. Structures at different axial positions therefore experience different sequences of excitation intensities. A fixed-focus camera records the fluorescence images while the plate, specimen and detection focus remain stationary. The phase plate is the only learned optical element; once trained, the same plate is used with each reconstruction method.

The design question is which measurement responses the plate should produce. Depth-sensitive optics have

been designed using Fisher information and response coherence [21, 25], while information-driven methods optimize measurements independently of a reconstruction decoder [14, 18]. We build on this principle for angle-scanned excitation. At each lateral position, we form the camera response of a voxel across illumination angles, including excitation, emission blur and pixel averaging. We learn the plate by minimizing weighted squared correlations between these signatures at different depths. The criterion compares finite axial alternatives through the complete measurement model, using neither specimen examples nor a reconstruction solver.

We evaluate the resulting encoder at both the measurement and reconstruction stages. A photon-count comparison tests whether learning makes neuronal references easier to distinguish from alternatives that preserve their lateral projection and mean depth. Reconstruction then uses total variation (TV), pretrained plug-and-play (PnP), and neuron-trained PnP under matched acquisition conditions. We report both axial-profile and centroid errors on simulated mouse pyramidal neuron measurements, and use controlled filament crossings to test separation and axial ordering. Objective comparisons and acquisition controls assess where the gains come from and the conditions under which they persist.

Our contributions are:

1. **Decoder-free excitation design.** We learn a fixed, defocused phase plate for angle-scanned fluorescence microscopy by reducing correlations between detector-integrated depth signatures, without specimen examples or a reconstruction decoder.
2. **Assessment of axial structure.** We evaluate the optical design through measurement discrimination, axial-profile and centroid accuracy, and controlled crossing separation.
3. **Recovery across reconstruction priors.** Paired comparisons demonstrate improvements with TV and two PnP configurations, alongside controls for alternative optical objectives, acquisition budgets, noise and optical registration.

2. Related work

Depth encoding in fluorescence microscopy. Three-dimensional fluorescence microscopy can encode depth through detection or excitation. Light-field microscopy, DiffuserCam and Miniscope3D reconstruct volumes from depth-dependent detection responses [1, 4, 25]. Excitation-based methods exploit illumination that varies with depth. Yeh *et al*. [26] combine translated speckle patterns with axial scanning, while Pascucci *et al*. [16] use axial speckle decorrelation for volumetric recovery from a two-dimensional scan, with saturation improving resolution. Angular scanning through a fixed scattering layer supports two-dimensional fluorescence recovery [27]; tilted pattern projection also produces depth-dependent lateral shifts [9]. DEEP and DEEP^2 use excitation patterning to recover fluorescence through scattering media [24, 29]. We study axial recovery in a non-scattering, linear-fluorescence model, learning a fixed, defocused phase plate for angular excitation scanning.

Learning optical encoders. Optical design can be guided by reconstruction performance or by properties of the measurements. End-to-end methods jointly optimize optics and reconstruction networks [8, 15, 22], while Kellman *et al*. [11] optimize illumination for a specified reconstruction algorithm. Decoder-independent criteria include Fisher information for 3D localization [21] and coherence between optical responses. Miniscope3D [25] designs a detection-side phase mask by penalizing cross-coherence between depth-dependent point-spread functions, including lateral shifts. Gram-matrix and coherence penalties also have precedents in sensing-matrix design [5, 6]. IDEAL and IDEAL-IO [14, 18] optimize estimated mutual information through fitted measurement-distribution models. Our criterion belongs to the coherence family and targets a specific excitation encoder: each signature collects a voxel's detector-integrated response across illumination angles, and the loss compares depths at a fixed lateral position. It requires no specimen examples, distribution fitting or reconstruction solver. We evaluate this design through both measurement discrimination and recovery across priors.

Volumetric reconstruction and priors. TV regularization promotes piecewise-smooth structure [20], while PnP incorporates denoisers as implicit priors [10, 23]. Multi-slice fusion applies lower-dimensional denoisers along complementary orientations of a shared volume [12]. These methods provide different prior assumptions for evaluating the same optical design. We keep the prior networks fixed across encoders and assess both axial-profile accuracy and separation of controlled crossing structures.

3. Decoder-free optical learning

3.1. Fixed-focus excitation encoding

Acquisition. For transverse position $\mathbf{r} = (x, y)$, illumination direction t produces the incident plane wave $u_t^{\mathrm{in}}(\mathbf{r}) = a_t e^{\mathrm{i}k_{\mathrm{ex}}\mathbf{s}_t^\top \mathbf{r}}$, where $\mathbf{s}_t$ is the transverse direction-cosine vector and $k_{\mathrm{ex}} = 2\pi/\lambda_{\mathrm{ex}}$ for refractive index one. The angular grid is specified in the supplement. The field passes through the fixed phase plate, $\tau(\mathbf{r}) = e^{\mathrm{i}\phi(\mathbf{r})}$, and propagates a distance D to the intermediate illumination image plane by angular-spectrum propagation. The objective pupil filters the modulated field before a Richards–Wolf relay [13, 19] produces the specimen excitation. We use a vectorial model for the high-NA relay, with x-polarized input and intensity $I_t = \sum_{c\in\{x,y,z\}} |E_{t,c}|^2$. Coordinates are in specimen-equivalent units. One fluorescence image is recorded per direction at a fixed detection focus; acquisition is sequential.

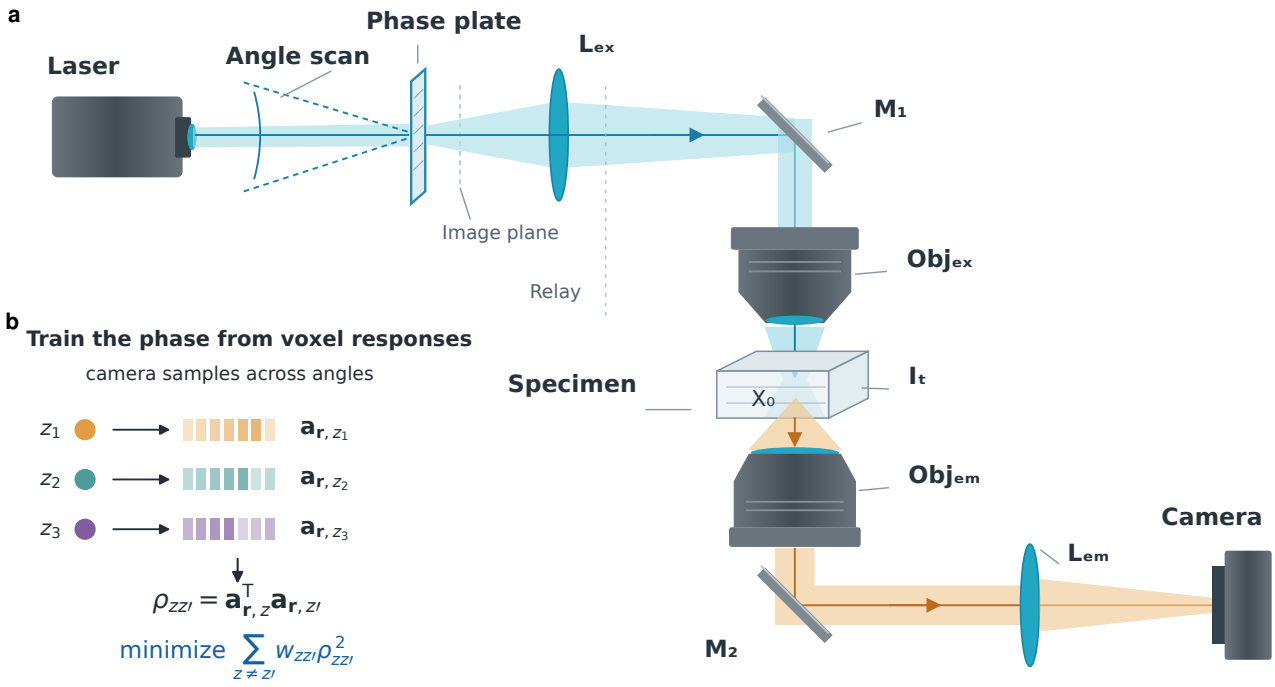


Figure 2. **Angle-scanned acquisition and decoder-free training.** (a) Only the incident angle changes; the phase plate, specimen and detection focus remain stationary. Blue: excitation; orange: fluorescence. (b) Schematic voxel responses across camera samples and illumination angles. Training compares depths at the same lateral position and updates the phase through the optical model.

Depth encoding. Propagation after the plate creates spatial intensity structure. Changing the incident angle shifts this structure laterally, with displacement increasing with propagation distance. In the ideal paraxial thin-screen model, a change $\Delta \mathbf{s}$ produces $\Delta \mathbf{r} \approx d\,\Delta \mathbf{s}$ after distance d. Since d varies through the specimen, different depths experience different excitation sequences during the scan. Plate defocus places the shear pivot outside the specimen. This relation explains the encoding principle; the high-NA vectorial model includes finite-pupil effects and departures from pure translation.

Image formation. Let $X_0(\mathbf{r}, z)$ denote fluorescence response per unit excitation. Under linear, unsaturated fluorescence, the emission source is proportional to $X_0 I_t$ [26]. For laterally shift-invariant detection, the noiseless camera-plane irradiance is

$$g_t(\mathbf{r}) = \int_\Omega h_{\mathrm{em}}(\mathbf{r}-\mathbf{r}', z_{\mathrm{focal}}-z') X_0(\mathbf{r}', z') I_t(\mathbf{r}', z')\, \mathrm{d}^2\mathbf{r}'\, \mathrm{d}z', \tag{1}$$

where h_{em} is the emission intensity PSF and Ω the specimen volume. Camera pixel p, occupying D_p, records $\hat{Y}^0_{t,p} = |D_p|^{-1} \int_{D_p} g_t(\mathbf{r})\, \mathrm{d}^2\mathbf{r}$. Fluorescence contributions add incoherently before pixel averaging. Stacking these samples over angles and pixels gives the noiseless vector $\bar{\mathbf{y}}$. Excitation mixes specimen spatial frequencies before detection blur, allowing the angle sequence to carry information that a single image does not resolve.

Voxel-integrated responses. The illumination can vary within a reconstruction voxel. We therefore evaluate the optics on a finer grid and integrate its responses over the reconstruction basis. If B is the fine-grid operator and U replicates each reconstruction coefficient over its fine-grid cells, then $\mathsf{A} = \mathsf{BU}$. This is exact for the discretized model restricted to a piecewise-constant specimen, while sub-voxel specimen variation remains an approximation error. The operator thus describes the camera response of a reconstruction voxel, rather than its response at a single point. Assembly and residual details are in the supplement.

3.2. Learning depth signatures

Training objective. For a sampled angle set $\mathcal{T}$, a voxel's signature is its column of A restricted to the camera samples recorded at those angles. It includes excitation modulation, emission blur and pixel averaging. At lateral position $\mathbf{r}$ and depth z, let $\mathbf{a}_{\mathbf{r},z}$ denote this signature after centering and unit-norm normalization. We optimize the phase through

$$\mathcal{L}(\phi) = \mathbb{E}_{\mathbf{r},\mathcal{T}} \frac{\sum_{z \neq z'} w_{zz'} \left(\mathbf{a}_{\mathbf{r},z}^\top \mathbf{a}_{\mathbf{r},z'}\right)^2}{\sum_{z \neq z'} w_{zz'}}. \tag{2}$$

Selected depth pairs receive weight two and other distinct pairs receive weight one; the pairing and sampling are specified in the supplement. Normalization prevents uniform amplification from reducing the loss. Gradients pass through the excitation and detection model. Optical training uses no specimen examples, fitted measurement distribution or reconstruction solver.

Interpretation. Stack the normalized signatures as columns of $\mathsf{S}_{\mathbf{r}}$ and write $\mathsf{G}_{\mathbf{r}} = \mathsf{S}_{\mathbf{r}}^{\top}\mathsf{S}_{\mathbf{r}}$. Matching one signature against this bank gives its column of $\mathsf{G}_{\mathbf{r}}$: the diagonal is one, and off-diagonal entries measure responses at other depths. Apart from constant normalization, the unweighted loss is the Gram penalty $\|\mathsf{G}_{\mathbf{r}} - \mathsf{I}\|_F^2$ [5, 6]. It compares finite axial alternatives across the sampled depth range. This motivates the criterion but does not guarantee inversion of a volume: responses from different lateral positions also interact. Reconstruction uses the original, uncentered operator A, and its performance is evaluated separately.

Centering a signature across angles and camera pixels differs from subtracting the mean of its angular excitation sequence. For a localized camera footprint, a common positive component can remain even when the angular fluctuations are weakly correlated. Learning can reduce recorded-response similarity by increasing angular contrast and making depths respond to different directions. The supplement derives this interpretation under a common-footprint approximation; the numerical model uses the full voxel responses.

4. Experimental setup

All experiments use the simulated microscope of Sec. 3, with linear fluorescence in a homogeneous, non-scattering medium. We use a common-wavelength approximation of 532 nm for excitation and emission, with numerical apertures of 0.8 and 0.1, respectively. The phase plate has $0.32\,\mu$m features and lies $6.6\,\mu$m before the intermediate illumination image plane, in specimen-equivalent coordinates. The full scan contains 256 directions. Optical fields use $0.08\,\mu$m lateral and $0.32\,\mu$m axial sampling, followed by $0.32\,\mu$m analysis sampling and $0.64\,\mu$m reconstruction voxels.

Volumes and compression. Reference volumes come from two-photon stacks of mouse pyramidal neurons in the public DEEP^2 dataset [24], min–max normalized and resampled to the analysis grid. Measurements are generated before averaging the specimen into reconstruction voxels. We reconstruct 16 axial planes of 48×48 voxels ($N = 36{,}864$), covering $30.7 \times 30.7 \times 10.2\,\mu\text{m}^3$ (lateral $\times$ lateral $\times$ axial). Each camera image has 12×12 samples. Acquiring $k = 256$, 64, or 32 directions gives measurement compression $C = N/(144k)$ of 1, 4, and 8. For $C > 1$, we evaluate five randomly selected direction subsets shared across encoders.

Dataset splits and validation. The test set contains six cells from three animals. Two crops per cell were prespecified by a structure score. After reconstruction parameters were fixed, eight more crops were randomly selected per cell, giving 60 test crops for the main comparison (Sec. 5.2). Other full-field experiments use the 12 prespecified crops unless stated otherwise. Neuron-trained denoisers use 459 crops from a fourth animal. Scalar reconstruction parameters are selected separately for each encoder and acquisition condition by validation PSNR on two vascular crops from another animal. Test volumes enter neither training nor parameter selection.

Reconstruction priors. We evaluate three methods, none trained jointly with the plate. *TV* uses box-constrained least squares with isotropic three-dimensional total variation [20], solved by ADMM [3]. *Pretrained PnP* applies a natural-image DnCNN [28] along the xy, yz and zx slices of a shared volume, with a sparsity branch, following multi-slice fusion [12]. *Neuron-trained PnP* uses a neuron-trained 2D slice network and an additional neuron-trained 3D branch. Both combine measurement consistency, denoising and sparsity through PnP-ADMM [10, 23]. Prior networks are frozen across encoders; scalar parameters follow the validation protocol above. Solver updates, constraint handling and denoiser training are given in the supplement.

Optical controls and dose. The main comparison pairs the learned plate with its random initialization. Two optimization seeds change the sampled positions and directions while retaining this initial phase. Additional random plates, a defocused amplitude mask, a plate-free scan and an in-focus DMD provide supporting controls. Excitation is simulated beyond the specimen field and normalized to a common mean over all 256 directions and the specimen volume. Counts follow $\text{Poisson}(\kappa_0\, 256\, \bar{y}_i/k)$ for k directions, where $\bar{y}_i$ is the noiseless measurement. Thus deposited excitation is matched exactly for the full scan and in expectation over random subsets. We call this matched nominal dose. For each specimen, κ_0 gives the random plate 1000 expected photons per pixel per image at the mean full-scan signal and is shared across encoders.

Axial-profile and image-quality metrics. We report centroid error for axial placement and Wasserstein-1 distance (W1) for the full axial intensity profile [17]. At each lateral column c, we normalize the non-negative reference and reconstructed profiles and form their cumulative distributions

F_c and $\hat{F}_c$. The primary profile error is

$$E_{\mathrm{W1}} = \frac{\Delta z}{\sum_{c\in\mathcal{S}} w_c} \sum_{c\in\mathcal{S}} w_c \sum_{\ell} \left| F_c(z_\ell) - \hat{F}_c(z_\ell) \right|, \quad (3)$$

where $\Delta z = 0.64\,\mu\mathrm{m}$, w_c is reference column brightness, and $\mathcal{S}$ contains columns exceeding 10 % of its maximum. W1 measures profile displacement in micrometers and can detect merging that preserves a centroid. Centroid error uses the same columns and weights. We also report PSNR and foreground SSIM, since column-normalized W1 does not measure brightness or lateral accuracy. Depth-blind references and metric details are in the supplement.

Measurement distinguishability. For each reference we construct a merged-depth alternative with the same lateral projection and column centroids. We compare their expected fine-grid photon measurements using Poisson KL, fitting the alternative's unknown global brightness. This decoder-free test uses the 59 crops containing layered columns. Exact alternatives, nuisance fitting and background variants are specified in the supplement.

Crossing structures. Success requires separating both depths at a filament crossing and correctly ordering the arms. The threshold is the smallest tested gap with at least 80 % success at each of two axial centers and all larger tested gaps, using three geometries and five draws. A merged control tests false splitting. Peak, ordering and phantom definitions are given in the supplement.

Statistics. Comparisons pair specimens, angular subsets, dose and noise protocol. We average draws within each crop before computing paired changes of means. The 95 % intervals use 10,000 bootstrap resamples of the six cells, retaining their crops together. KL ratios are summarized geometrically. The 60 crops are repeated samples from six cells, rather than 60 independent biological specimens.

5. Results

5.1. Learning depth-discriminative responses

We first test the measurements before applying a reconstruction prior. At the preset field-center location, median angular contrast (standard deviation/mean over directions) rises from 0.58 to 1.53, while cosine similarity between the uncentered angular responses falls from 0.83 to 0.41 (Fig. 3a). These values describe one preset position. Over the interior field, mean absolute depth-signature correlation falls from 0.84 to 0.48 (Fig. 3b).

For each eligible neuronal crop, the merged-depth alternative preserves its lateral projection and every column's centroid. Relative to the random plate, the learned plate increases brightness-profiled Poisson KL to this alternative by a geometric mean factor of 10.0 (95 % CI: 8.7–11.8), with an increase in all 59 crops (Fig. 3c). Added background gives a factor of 10.4. The measurements therefore separate these selected alternatives more strongly, even though their mean depths agree. This diagnostic is distinct from reconstruction accuracy and does not establish global inverse stability.

5.2. Recovering neuronal volumes across priors

At $C = 8$, the learned plate reduces mean axial W1 by 41 % with TV, 50 % with pretrained PnP and 28 % with neuron-trained PnP (Tab. 1). All cell-bootstrap intervals exclude zero, and every crop improves under each prior. The 48 randomly selected crops were added after reconstruction parameters were fixed, supporting the result beyond the 12 prespecified crops. A second optimization seed gives similar reductions of 41 %, 50 % and 29 % from the same initial phase.

Centroid error falls by 30–54 %, showing improved axial placement alongside the profile gains, and PSNR increases by 1.6–1.8 dB (Tab. 1). Figure 4 shows axial sections from three ranked examples. The learned plate reduces visible smearing in the larger-gain case, while differences are smaller in the median and lower-gain sections. These neuronal views illustrate profile accuracy, not separation of overlapping neurites; the next experiment tests separation directly.

The optical and prior gains are complementary. Neuron-trained PnP has the lowest random-plate W1, yet still benefits from learning the illumination. In a descriptive cross-prior comparison, learned-plate TV gives lower W1 than random-plate neuron-trained PnP: 0.218 versus $0.271\,\mu\mathrm{m}$, a 20 % reduction (95 % CI: 16–23 %). This specific contrast shows that improving the measurements can rival a change of prior; it does not rank conventional and learned reconstruction methods generally.

5.3. Separating crossing structures

The crossing test asks whether improved depth responses yield two separated structures in the correct axial order (Fig. 5). With TV and pretrained PnP, the smallest tested gap meeting this criterion falls from 3.84 to $2.56\,\mu\mathrm{m}$. This holds at both axial centers, including the one outside the weighted mirror pairing used during training. Neuron-trained PnP retains a $3.84\,\mu\mathrm{m}$ threshold, although success rates increase at smaller gaps. Thus, lower profile error does not necessarily change the smallest gap resolved by a decoder.

Neither plate falsely splits the merged control. Both remain unreliable for closely stacked three-filament structures; unequal-intensity and three-filament results are in the supplement.

	axial W1 (μm) ↓		centroid error (μm) ↓		PSNR (dB) ↑		W1 reduction
prior	random	learned	random	learned	random	learned	[95 % CI]
TV	0.371	0.218	0.259	0.150	36.70	38.29	41 % [38, 45]
pretrained PnP	0.579	0.290	0.490	0.224	35.72	37.56	50 % [47, 53]
neuron-trained PnP	0.271	0.195	0.197	0.139	38.01	39.60	28 % [26, 29]

Table 1. **Effect of optical learning with fixed reconstruction priors.** Mean over 60 neuronal crops at eightfold measurement compression and matched nominal excitation dose. W1, centroid error and PSNR use the same reconstructions. Draws are averaged within each crop before averaging crops; W1-reduction intervals use paired cell-bootstrap resampling.

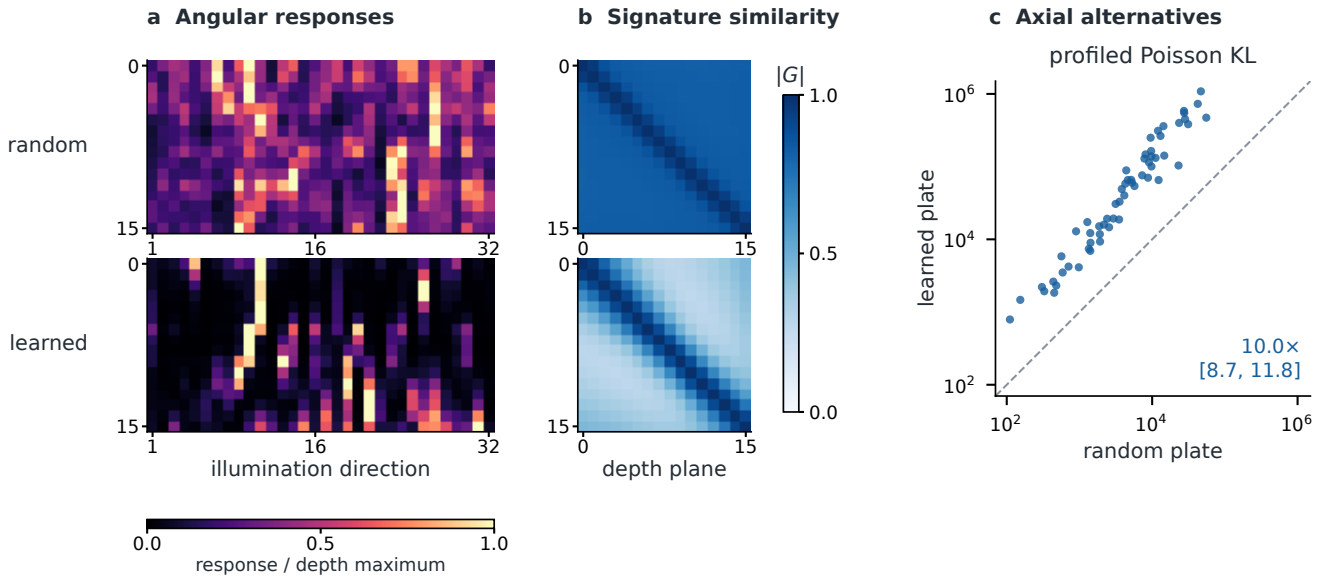


Figure 3. **Optical learning changes the angular depth responses.** (a) Responses on each voxel's own camera pixel at the preset field center, with each depth scaled by its maximum. (b) Absolute full-signature correlations, centered over angles and camera pixels as in the training loss, then averaged over interior lateral positions. (c) Brightness-profiled Poisson KL to a merged-depth alternative with the same projection and column centroids. Points average five subsets per eligible crop; the dashed line denotes equal KL.

5.4. Comparing optical training objectives

Our loss aggregates correlations between finite depth pairs. The coherence alternative emphasizes the most similar pairs, while the depth-Fisher surrogate rewards local axial sensitivity. We compare these implemented criteria under matched initialization, sampling and optimization budgets (Fig. 6). All three improve upon the random plate; the proposed objective gives 13–20 % lower W1 than the alternatives across both optimization seeds and all three priors. An uncentered ablation performs similarly under neuron-trained PnP, so the gain cannot be attributed to centering. Exact losses, absolute errors and the $C = 1$ comparison are in the supplement.

5.5. Measurement and dose trade-offs

Learning also changes the acquisition trade-off. At 0.3 times the reference excitation dose, the learned plate has lower W1 than the random plate at full dose under all three priors: reductions are 26 % [20, 33], 39 % [34, 45] and 22 % [16, 28] for TV, pretrained PnP and neuron-trained PnP (Tab. 2). This comparison reduces excitation while retaining 32 images.

prior	random 1× dose	learned 0.3× dose	reduction [95 % CI]
TV	0.318	0.234	26 % [20, 33]
pretrained PnP	0.496	0.302	39 % [34, 45]
neuron-trained PnP	0.282	0.220	22 % [16, 28]

Table 2. **Recovery at reduced excitation dose.** Mean W1 (μm, $C = 8$, 12 prespecified crops). Reductions compare learned at 0.3× with random at 1× dose, with cell-bootstrap 95 % intervals.

At matched nominal dose, 32 learned-plate images yield lower W1 than 256 random-plate images under TV and pretrained PnP. Neuron-trained PnP gives means of 0.206 and 0.203 μm, respectively. Fewer images receive longer exposures, so this result concerns measurement count rather

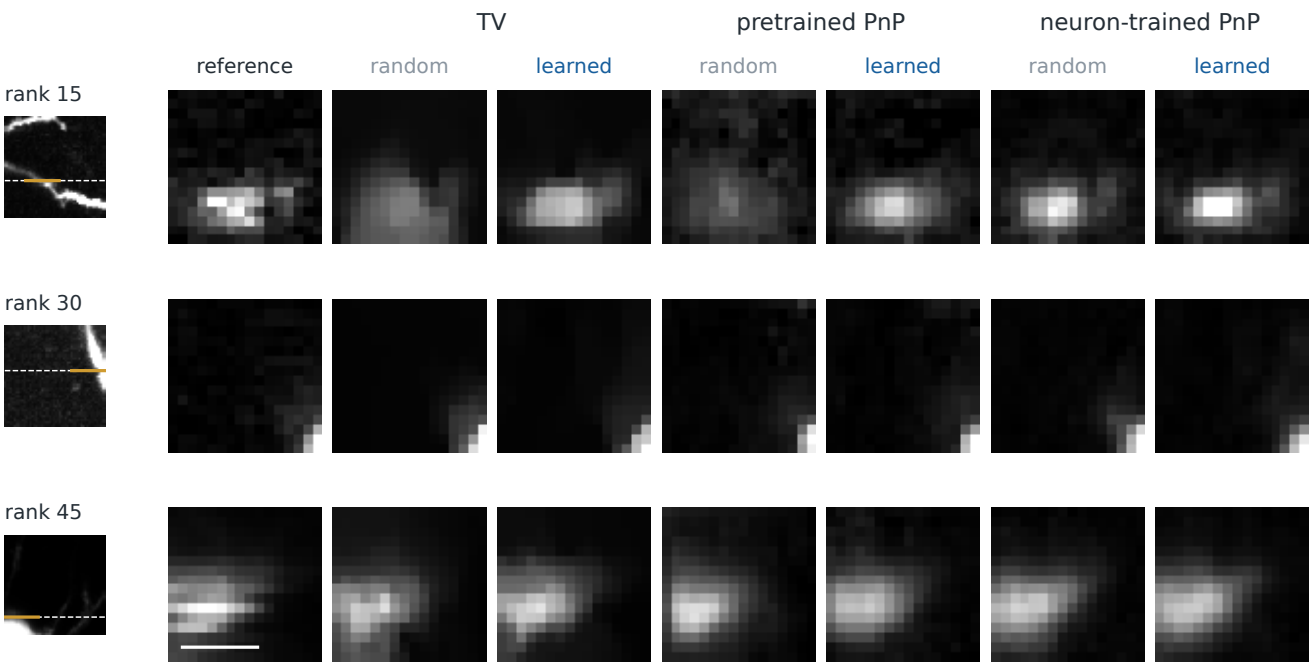


Figure 4. **Neuronal axial sections across reconstruction priors.** Rows are ranks 15, 30 and 45 of 60 by mean fractional W1 change across priors after averaging draws; rank 1 has the largest reduction. Locators mark the section and zoom window. The horizontal axis is lateral position; depth increases downward from the first plane. Each $10.2 \times 10.2\,\mu\mathrm{m}$ section uses the same 16-voxel lateral window around the selected profile column, shifted inward at crop boundaries. Intensities share a reference-derived linear range within each row ($C = 8$, draw 0, matched nominal dose; bar: $5\,\mu\mathrm{m}$). Registered profiles and selection details are in the supplement.

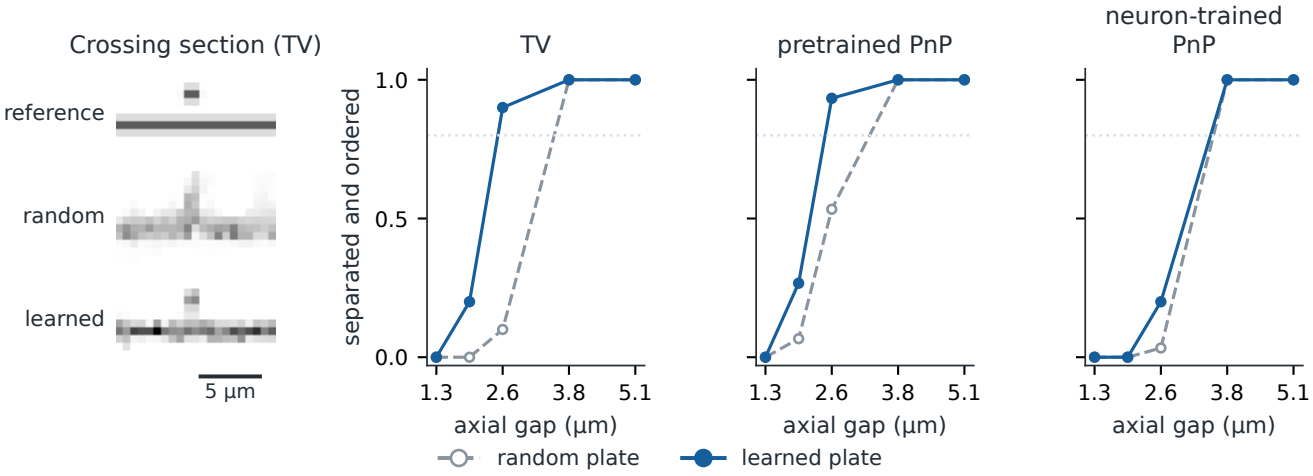


Figure 5. **Testing separation and axial ordering.** Left: TV sections through a two-filament crossing at a $2.56\,\mu\mathrm{m}$ axial gap, on a common intensity scale (bar: $5\,\mu\mathrm{m}$). Right: success rates pooled across geometries, axial centers and draws. The 80 % criterion is checked separately at each center and all larger tested gaps. The smallest qualifying gap falls from 3.84 to $2.56\,\mu\mathrm{m}$ with TV and pretrained PnP; neuron-trained PnP retains $3.84\,\mu\mathrm{m}$.

than an eightfold reduction in acquisition time. Complete dose and compression curves are in the supplement.

5.6. Noise and optical calibration

With background and read noise, the learned plate reduces W1 by 30 %, 53 % and 27 % relative to the random plate under the same conditions. Gains also persist when both

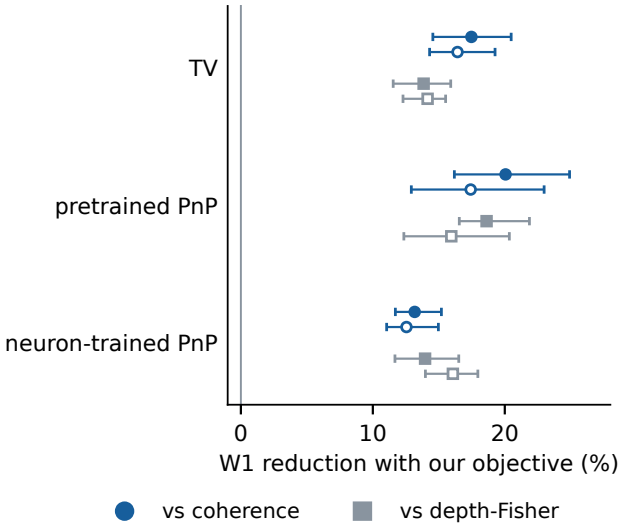


Figure 6. **Comparing decoder-free training criteria.** Paired W1 reductions with our plate relative to the implemented coherence and depth-Fisher alternatives ($C = 8$, 12 prespecified crops). Training shares the initial phase, sampling and optimization budget. Filled and open markers denote seeds 0 and 1, which change the sampled positions and angles. Bars show cell-bootstrap 95 % intervals.

encoders receive one tenth of the reference dose.

Registration errors distinguish encoder quality from model accuracy. Under neuron-trained PnP, plate-distance offsets of $\pm 0.5\,\mu$m retain reductions of 27–28 % when the operator is rebuilt using the known simulated offset and parameters are reselected on validation data. The phase plate is not retrained. Keeping the nominal operator instead removes the advantage under axial or lateral displacement: a one-feature lateral shift makes W1 11 % worse than random (95 % CI: 2–23 %). Uncalibrated pointing errors of 0.005–0.01 in direction cosine retain smaller reductions of 13–20 %. These results highlight the importance of characterizing the realized optical responses.

6. Discussion and limitations

Optical design across priors. Optimizing depth signatures improves recovery under both conventional and learned regularization. The gain with TV establishes that a neuron-trained prior is not required to benefit from the learned illumination. Neuron-trained PnP still improves when paired with the same plate, so optical design and a stronger prior provide complementary gains. The evaluation covers TV and two related PnP configurations. A comparison with end-to-end optical learning remains open.

Axial structure and separation. Here, axial structure is assessed through intensity profiles and controlled separation of crossing filaments. Centroid and W1 improvements agree, while crossing separation improves under two priors and remains unchanged under neuron-trained PnP. These outcomes show why accurate depth placement, profile recovery and separation should be assessed separately. The loss compares depths at one lateral position rather than all possible voxel pairs, and it does not guarantee unique recovery of neuronal branches or connections.

Physical scope. The study simulates linear fluorescence in a homogeneous, non-scattering medium with broad-depth-of-field detection and idealized components. Scattering and aberrations can change the optical responses, while saturation violates linear image formation. More selective excitation at a fixed mean dose also raises the peak intensity, making this assumption relevant to physical implementation. Registration tests show that an operator updated for a known displacement can retain gains, while nominal-model reconstruction can lose them. A physical implementation therefore requires characterization of the realized illumination and detection responses; estimating that calibration and validating the design experimentally remain open.

Evidence scope. The main comparison uses 60 crops from six cells across three animals. Cell-bootstrap intervals are approximate with six cells and do not establish population-level transfer across animals. Objective, dose and registration studies use the 12 prespecified crops. Both optical optimization seeds start from the same phase, so their agreement tests sampling variability rather than robustness to independent initial designs. The reported gains apply to the tested optical and specimen regime.

7. Conclusion

We demonstrate decoder-free optical learning for fixed-focus fluorescence microscopy by optimizing depth-dependent measurement signatures. The resulting phase plate improves axial-profile recovery across TV and two PnP methods, reducing W1 by 28–50 % and centroid error by 30–54 % in neuronal simulations. Controlled crossings show smaller recovered separations with two of the priors. These results support learning the excitation measurements as a way to improve axial recovery without tailoring the optical design to a reconstruction network.

LLM Usage Statement

LLM tools assisted with manuscript drafting, editing and figure-generation code.

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